

Quantitative Hasse-Principle Failures in the Fibers of a Fixed Keller Map

Roy van Rijn
Independent researcher

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Abstract

The classical polynomial $(X^3 - 19)(X^2 + X + 1)$ has a root over every completion of $\mathbb{Q}$, but no rational root, and at first looks like an isolated arithmetic curiosity. Starting from this Hasse failure, we ask whether the map, rather than only the fiber, can be kept fixed. The decisive observation is that after centering the quadratic factor, the whole family $(X^3 - a)(X^2 + X + 1)$ moves in exactly the two coefficients controlled by one quadratic-gauge target line.

This produces one explicit polynomial map $\Phi : \mathbb{A}_{\mathbb{Q}}^3 \rightarrow \mathbb{A}_{\mathbb{Q}}^3$ with Jacobian determinant one and geometric degree five. For every noncube integer $a > 1$ such that

$$a \equiv 1 \pmod{9}, \quad p \mid a \implies p \equiv 1 \pmod{3},$$

the map has a full fiber

$$\operatorname{Spec} \mathbb{Q}[X] / ((X^3 - a)(X^2 + X + 1)).$$

Every such fiber is reduced, has points over $\mathbb{R}$ and every $\mathbb{Q}_p$, and has no $\mathbb{Q}$ -point. The targets lie on a fixed rational line and have height $32a$. The local argument does not require a to be prime: its congruence-and-support core is multiplicatively closed, after which the perfect cubes are removed. A character-twisted Selberg–Delange argument then shows that the number of constructed targets of height at most B is asymptotic to

$$\frac{G_3(1)}{96\sqrt{\pi}} \frac{B}{\sqrt{\log B}},$$

where $G_3(1) > 0$ is an explicit Euler product. Finally, the classical degree bound for intersective polynomials shows that geometric degree five is best possible. Thus one exceptional quintic becomes a quantitatively large family of failures on a single map of the smallest possible geometric degree.

1 Introduction

There is a particularly sharp way for a local-global principle to fail. Take a finite étale scheme X over $\mathbb{Q}$. It has no singularities, no positive-dimensional families of points, and no geometry in which a rational point can hide. Nevertheless it may satisfy

$$X(\mathbb{R}) \neq \emptyset, \quad X(\mathbb{Q}_p) \neq \emptyset \text{ for every prime } p, \quad X(\mathbb{Q}) = \emptyset. \quad (1.1)$$

For a monic polynomial this says something completely concrete: at each completion at least one factor has a root, although none has a rational root. Such polynomials are called intersective.

The classical example of Berend and Bilu is

$$f_{19}(X) = (X^3 - 19)(X^2 + X + 1). \quad (1.2)$$

The two factors divide the local work between them. At primes $p \equiv 2 \pmod{3}$, cubing is invertible and the cubic supplies a root. At primes $p \equiv 1 \pmod{3}$, the quadratic supplies a cube root of unity. Only 2, 3, and 19 require separate attention. Globally, neither factor has a rational root.

The companion prescribed-fiber theorem turns every finite étale algebra of rank at least three into a full fiber of a Jacobian-one map of affine three-space [4]. In particular, it turns (1.2) into a degree-five Hasse failure. At first sight this seems to settle the Keller-map version of the problem.

It settles only the first quantifier order. The prescribed-fiber construction begins with a polynomial and builds a map around it. If the polynomial changes, the map generally changes as well. The question that remained was therefore:

Can one polynomial Keller map contain infinitely many Hasse-failing complete fibers?

This is a different problem. We no longer want to compile one arithmetic algebra into one map. We need an entire arithmetic family to fit inside the target pencil of a single map. That small change in quantifiers is where the problem becomes interesting.

Finding the line

The Berend–Bilu example already suggests the family

$$f_a(X) = (X^3 - a)(X^2 + X + 1). \quad (1.3)$$

For primes $a = \ell \equiv 1 \pmod{9}$, the same local proof works uniformly. But the expanded polynomial

$$f_a(X) = X^5 + X^4 + X^3 - aX^2 - aX - a$$

does not look like a one-coordinate motion in the usual linear inverse pencil: three coefficients move at once.

The useful step is almost embarrassingly small. Center the quadratic:

$$X = S - \frac{1}{2}, \quad X^2 + X + 1 = S^2 + \frac{3}{4}. \quad (1.4)$$

Now the parameter enters through

$$-a \left(S^2 + \frac{3}{4} \right),$$

so only the quadratic and constant coefficients move. Those are exactly the two target coefficients in the quadratic-gauge inverse equation

$$E_{\Pi, B, C}(S) = G_{\Pi}(S) - \frac{g_1}{2}(BS^2 + C). \quad (1.5)$$

Once (1.4) is made, the family (1.3) is no longer a search problem. It is a straight line in the target of one fixed map. This was the point at which the isolated example began to look like a family.

This gives the geometric part of the discovery. The arithmetic part then becomes unexpectedly generous. Primality of a is not used in the local argument. What is really needed is

$$a \equiv 1 \pmod{9}, \quad a \notin \mathbb{Q}^3, \quad p \mid a \implies p \equiv 1 \pmod{3}. \quad (1.6)$$

Thus the prime progression sits inside a much larger multiplicatively generated family. Its congruence-and-support core is a semigroup; the noncube condition removes the perfect cubes, so $\mathcal{A}$ itself is not literally closed under multiplication. The prime family has order $B/\log B$ by height; the full sufficient family has order $B/\sqrt{\log B}$.

The result

For $a > 1$, define

$$y_a = \left(-1, \frac{32a}{9}, \frac{8a+1}{3} \right) \in \mathbb{A}^3(\mathbb{Q}) \quad (1.7)$$

and let

$$\mathcal{A} = \left\{ a > 1 : \begin{array}{l} a \equiv 1 \pmod{9}, \ a \notin \mathbb{Q}^3, \\ p \mid a \implies p \equiv 1 \pmod{3} \end{array} \right\}. \quad (1.8)$$

Since a is an integer, the condition $a \notin \mathbb{Q}^3$ simply excludes perfect cubes.

Theorem 1.1 (Fixed-map theorem). *There is an explicit polynomial map*

$$\Phi : \mathbb{A}_{\mathbb{Q}}^3 \longrightarrow \mathbb{A}_{\mathbb{Q}}^3, \quad \det D\Phi = 1, \quad \text{gdeg}(\Phi) = 5,$$

such that, for every $a \in \mathcal{A}$,

$$\boxed{\Phi^{-1}(y_a) \simeq \text{Spec } \mathbb{Q}[X] / ((X^3 - a)(X^2 + X + 1))}. \quad (1.9)$$

This is a full reduced fiber. It has a point over $\mathbb{R}$ and over every $\mathbb{Q}_p$, but it has no $\mathbb{Q}$ -point.

The targets have primitive projective coordinates

$$[1 : y_a] = [9 : -9 : 32a : 24a + 3], \quad H(y_a) = 32a. \quad (1.10)$$

They are therefore pairwise distinct. The first parameter is $a = 19$, so the classical example is the first point on the line rather than an isolated construction. The old example has not disappeared; it has found its natural place.

Theorem 1.2 (Quantitative theorem). *Let*

$$M_{\Phi}(B) = \#\{a \in \mathcal{A} : H(y_a) \leq B\}.$$

Then

$$\boxed{M_{\Phi}(B) \sim \frac{G_3(1)}{96\sqrt{\pi}} \frac{B}{\sqrt{\log B}}}, \quad (1.11)$$

where

$$G_3(1) = \left[\frac{2}{3} L(1, \chi_{-3}) \prod_{p \equiv 2 \pmod{3}} (1 - p^{-2}) \right]^{1/2} > 0. \quad (1.12)$$

Consequently the total number of Hasse-failing rational targets of Φ of height at most B is at least of order $B/\sqrt{\log B}$.

The theorem counts the displayed sufficient family exactly; it does not claim an asymptotic for all Hasse-failing targets of Φ . This distinction matters. The bounded experiment later in the paper checks the family and the numerical convergence, but the asymptotic comes from the Euler product and Selberg–Delange.

The rest of the paper follows the route by which the construction becomes visible. Section 2 starts from the scalar inverse equation, writes down the fixed map, and proves that no boundary point is missing from the fiber. Section 3 lets the two factors in (1.3) divide the completions between them. Section 4 first counts the prime progression and then the larger multiplicative family. Section 5 returns to the classical degree obstruction and shows that five is not only attained, but forced.

2 The fixed map and its complete fibers

The construction is easiest to understand backwards. We want the inverse problem over a target to be a single quintic equation in a marked root S , and we want the parameter a to enter only through the coefficients of S^2 and 1. The quadratic gauge was designed precisely so that these two coefficients are target coordinates, while the derivative of the inverse equation cancels out of the Jacobian.

The centered identity from the introduction determines the seed. Put

$$G(S) = S^5 - \frac{3}{2}S^4 + \frac{3}{2}S^3 - \frac{5}{4}S^2 + \frac{9}{16}S. \quad (2.1)$$

The coefficients may look tailored, because they are: G is the fixed part left after translating $X^3(X^2 + X + 1)$ by $X = S - \frac{1}{2}$.

We first write the natural determinant-minus-two gauge map, and then make a small linear normalization to obtain Jacobian one. For source coordinates (x, y, z) , set

$$t = 1 + xy, \quad q = t^2 z + \frac{3}{8}y^2(1 + 3t), \quad (2.2)$$

and define $F_0 = (\Pi, B, C)$ by

$$\begin{aligned} \Pi &= tq, \\ B &= y + 8xq - \frac{40}{9}tq - \frac{32}{3}t^2x^2q^4 + \frac{80}{9}t^2x^3q^5, \\ C &= x(5 - 3t) - \frac{8}{3}x^3z + \frac{16}{3}(xq)^4 - \frac{16}{3}(xq)^5. \end{aligned} \quad (2.3)$$

These are polynomial coordinates over $\mathbb{Q}$. The formula is long, but there is no search hidden inside it: it is the coefficientwise polynomial pullback of the one-variable inverse equation below.

The determinant -2 is the natural normalization of the inverse equation. It is not an arithmetic restriction. Let

$$A(x, y, z) = \left(\frac{x}{2}, y, z\right), \quad L(\Pi, B, C) = (-\Pi, B, C), \quad (2.4)$$

and take as our fixed map

$$\boxed{\Phi = L \circ F_0 \circ A.} \quad (2.5)$$

Equations (2.2)–(2.5) are an explicit polynomial formula for Φ . The normalization has been chosen so that it does not increase the height of the target family.

2.1 Inverse equation

The inverse problem for (2.3) is one-dimensional. For an independent target coordinate Π , define

$$G_\Pi(S) = \frac{9}{16}S + \Pi \left(-\frac{5}{4}S^2 + \frac{3}{2}S^3 \right) - \frac{3}{2}\Pi^4S^4 + \Pi^5S^5 \quad (2.6)$$

and

$$E_{\Pi, B, C}(S) = G_\Pi(S) - \frac{9}{32}(BS^2 + C). \quad (2.7)$$

Proposition 2.1 (Quadratic-gauge specialization). *The map F_0 satisfies*

$$\det DF_0 = -2, \quad \text{gdeg}(F_0) = 5. \quad (2.8)$$

If $\pi \neq 0$ and $E_{\pi, b, c}$ is squarefree, then there is a natural fiber isomorphism

$$F_0^{-1}(\pi, b, c) \simeq \text{Spec } \mathbb{Q}[S]/(E_{\pi, b, c}). \quad (2.9)$$

Proof. We recall the short inverse-coordinate calculation, both to make the fiber claim self-contained and to account for every source point. On $t \neq 0$ put

$$S = \frac{x}{t}, \quad Q = y + xq, \quad D = 1 - SQ + \Pi S^2. \quad (2.10)$$

Since $\Pi = tq$, one has $D = t^{-1}$. With

$$\beta(\Pi, S) = \frac{G'_\Pi(S)/(9/16) - 1 - \Pi S^2}{S}, \quad (2.11)$$

direct expansion of (2.3) gives

$$B = Q + \beta(\Pi, S), \quad C = \frac{32}{9}G_\Pi(S) - BS^2. \quad (2.12)$$

The second identity is precisely $E_{\Pi, B, C}(S) = 0$, and differentiation gives

$$D = \frac{16}{9}E'_{\Pi, B, C}(S). \quad (2.13)$$

At fixed Π , the Jacobian of $(S, Q) \mapsto (B, C)$ is $-2D$; the Jacobian of the source chart $(x, y, z) \mapsto (\Pi, S, Q)$ is D^{-1} . Their product is -2 , proving the first identity in (2.8).

If a source point lies over (π, b, c) with $\pi \neq 0$, then $tq = \pi$, so t and q are units and the chart above contains the entire fiber. Conversely, let s be a root of $E_{\pi, b, c}$ over a commutative $\mathbb{Q}$ -algebra and suppose $E'_{\pi, b, c}(s)$ is a unit. Put

$$\begin{aligned} d &= \frac{16}{9}E'_{\pi, b, c}(s), & Q &= b - \beta(\pi, s), & t &= d^{-1}, \\ x &= sd^{-1}, & y &= Q - \pi s, & q &= \pi d, \\ z &= d^2 \left(q - \frac{3}{8}y^2(1 + 3t) \right). \end{aligned} \quad (2.14)$$

Equations (2.10)–(2.13) show directly that this reconstructs a unique source point. If $E_{\pi, b, c}$ is squarefree, Bézout's identity for E and E' makes $E'(s)$ a unit over every test algebra. Hence (2.14) is a natural inverse and proves (2.9).

Finally, the generic polynomial (2.7) has degree five in S and is irreducible over $\mathbb{Q}(\Pi, B, C)$. Indeed, over $\mathbb{Q}(\Pi, B)[C]$ it is primitive and linear in C ; in a nontrivial factorization one factor would be independent of C , and comparison of the coefficient of C would make that factor divide a nonzero scalar. The reconstruction identifies the actual source field with the field obtained by adjoining S . Thus its degree is five, proving the second identity in (2.8). $\square$

Since the determinants of A and L are $1/2$ and -1 , respectively, (2.8) gives

$$\det D\Phi = (-1)(-2) \left(\frac{1}{2} \right) = 1, \quad \text{gdeg}(\Phi) = 5. \quad (2.15)$$

2.2 Why the target line appears

At this point the map is fixed. What remains is to see the whole intersective family inside its target. Set $\Pi = 1$. For $a \in \mathbb{Q}$, put

$$\tilde{y}_a = \left(1, \frac{32a}{9}, \frac{8a+1}{3} \right). \quad (2.16)$$

The elementary identity

$$\left(S - \frac{1}{2}\right)^3 \left(S^2 + \frac{3}{4}\right) = G(S) - \frac{3}{32} \quad (2.17)$$

gives

$$E_{\tilde{y}_a}(S) = \left(\left(S - \frac{1}{2}\right)^3 - a \right) \left(S^2 + \frac{3}{4}\right). \quad (2.18)$$

After $X = S - \frac{1}{2}$, this becomes

$$(X^3 - a)(X^2 + X + 1). \quad (2.19)$$

The target line is now visible. Changing a changes the target, not the map. The centered quadratic has no linear term, so the parameter is absorbed exactly by B and C .

The target y_a in (1.7) is $L(\tilde{y}_a)$. The source automorphism A and Proposition 2.1 therefore give (1.9) whenever (2.19) is squarefree.

Its two factor discriminants and their resultant are

$$\text{disc}(X^3 - a) = -27a^2, \quad \text{disc}(X^2 + X + 1) = -3, \quad \text{Res} = (a - 1)^2. \quad (2.20)$$

Thus every $a > 1$ in $\mathcal{A}$ gives a reduced fiber of rank five. By (2.15), the fiber is full.

3 The local-global calculation

The geometric work is now finished: the entire fiber is the root scheme of one displayed polynomial. The remaining argument is arithmetic, and it is here that the factorization reveals why the family is so flexible. The cubic and quadratic do not both need a root at every place. They only need to cover all places between them.

Fix $a \in \mathcal{A}$ and write

$$f_a(X) = (X^3 - a)(X^2 + X + 1). \quad (3.1)$$

The cubic has no rational root because a is not a cube, while the quadratic is irreducible because its discriminant is -3 . Hence f_a has no rational root. The fiber description (1.9) then gives

$$\Phi^{-1}(y_a)(\mathbb{Q}) = \emptyset. \quad (3.2)$$

We now let the two factors divide the completions according to the residue class of the prime.

The real place. The polynomial $X^3 - a$ has the real root $\sqrt[3]{a}$.

The prime 2. Every prime divisor of a is $1 \pmod{3}$, so a is odd. Thus $X = 1$ is a root of $X^3 - a$ modulo 2, and its derivative $3X^2$ is a 2-adic unit. Hensel's lemma supplies a root in $\mathbb{Z}_2$.

The prime 3. Write $a = 1 + 9k$ and substitute $X = 1 + 3T$. Then

$$X^3 - a = 9h(T), \quad h(T) = T + 3T^2 + 3T^3 - k. \quad (3.3)$$

Modulo 3,

$$h(T) \equiv T - k, \quad h'(T) \equiv 1. \quad (3.4)$$

The solution $T \equiv k \pmod{3}$ lifts to a root in $\mathbb{Z}_3$.

All other primes. Let $p \neq 2, 3$. If $p \equiv 1 \pmod{3}$, then $\mathbb{F}_p^\times$ contains a nontrivial cube root of unity. It is a simple root of $X^2 + X + 1$, so the quadratic has a root in $\mathbb{Q}_p$. This case includes every prime dividing a .

If $p \equiv 2 \pmod{3}$, then $p \nmid a$ and cubing is an automorphism of $\mathbb{F}_p^\times$. The polynomial $X^3 - a$ therefore has a nonzero root modulo p . Its derivative is nonzero, so the root lifts to $\mathbb{Q}_p$.

This exhausts all places and proves the local assertions of Theorem 1.1. Notice that the same two factors alternate their role with the residue class of p . The condition $a \equiv 1 \pmod{9}$ is needed only at 3; the prime-support condition controls the primes dividing a . This is why primality disappears from the theorem: the local proof remembers only a congruence and the support of a . That is the arithmetic surprise behind the larger family.

Remark 3.1. *The first prime parameter is $a = 19$, giving the classical Berend–Bilu algebra (1.2). The first broader composite example is $a = 91 = 7 \cdot 13$. The construction is therefore not merely a repackaging of primes in one arithmetic progression.*

Remark 3.2. *The reducibility of f_a is forced by the local condition. A connected finite étale $\mathbb{Q}$ -scheme of degree greater than one has a transitive Galois action containing a derangement. Chebotarev then gives an unramified prime at which the scheme has no point. Hasse-failing zero-dimensional schemes must consequently have at least two arithmetic components; see Sonn [3] for the corresponding subgroup-covering formulation.*

4 Counting the multiplicative family

The first infinitude proof is immediate: take $a = \ell$ prime with $\ell \equiv 1 \pmod{9}$, and invoke Dirichlet. That already puts infinitely many Hasse failures on one map. But it throws away most parameters allowed by the local calculation. It proves infinitude, but it does not yet reveal the natural size of the family.

The support condition in (1.8) is multiplicative. Products of suitable primes remain suitable, provided their residue modulo 9 returns to 1, and the number of available prime factors is unbounded. The correct scale is therefore not the prime-counting scale $B/\log B$, but a Selberg–Delange scale. We first translate parameter size into target height and then count the semigroup.

We use the standard height induced by

$$\mathbb{A}^3(\mathbb{Q}) \longrightarrow \mathbb{P}^3(\mathbb{Q}), \quad (u, v, w) \longmapsto [1 : u : v : w]. \quad (4.1)$$

For $a \in \mathcal{A}$, equation (1.7) gives

$$[1 : y_a] = [9 : -9 : 32a : 24a + 3]. \quad (4.2)$$

Because $a \equiv 1 \pmod{9}$, these coordinates are primitive; for $a > 1$, their maximum absolute value is $32a$. Hence

$$H(y_a) = 32a. \quad (4.3)$$

4.1 The full sufficient family

Ignore the congruence and cube conditions for a moment, and count all positive integers supported on primes $1 \pmod{3}$. Their Dirichlet series records the density $1/2$ of the allowed primes:

$$\mathcal{F}_3(s) = \prod_{p \equiv 1 \pmod{3}} (1 - p^{-s})^{-1} = \zeta(s)^{1/2} G_3(s). \quad (4.4)$$

If χ_{-3} is the nonprincipal character modulo 3, comparison of Euler factors gives

$$G_3(s) = \left[L(s, \chi_{-3})(1 - 3^{-s}) \prod_{p \equiv 2 \pmod{3}} (1 - p^{-2s}) \right]^{1/2}. \quad (4.5)$$

The branch is chosen positive for real $s > 1$. In particular,

$$G_3(1) = \left[\frac{2}{3} L(1, \chi_{-3}) \prod_{p \equiv 2 \pmod{3}} (1 - p^{-2}) \right]^{1/2} > 0. \quad (4.6)$$

The Selberg–Delange theorem applied to (4.4) yields

$$\#\{n \leq X : p \mid n \Rightarrow p \equiv 1 \pmod{3}\} \sim \frac{G_3(1)}{\sqrt{\pi}} \frac{X}{\sqrt{\log X}}. \quad (4.7)$$

Such integers lie in the order-three subgroup

$$H = \{1, 4, 7\} \subset (\mathbb{Z}/9\mathbb{Z})^\times.$$

Here is the character step explicitly. If ψ is a character of H , put

$$\mathcal{F}_{3,\psi}(s) = \prod_{p \equiv 1 \pmod{3}} (1 - \psi(p)p^{-s})^{-1}.$$

The prime number theorem in the three reduced classes 1, 4, 7 (mod 9) shows that its Selberg–Delange exponent is

$$\rho_\psi = \frac{1}{6} \sum_{r \in H} \psi(r).$$

Thus $\rho_\psi = 1/2$ for the trivial character and $\rho_\psi = 0$ for either nontrivial character. The character-twisted Selberg–Delange theorem therefore gives

$$\sum_{\substack{n \leq X \\ p \mid n \Rightarrow p \equiv 1 \pmod{3}}} \psi(n) = o\left(\frac{X}{\sqrt{\log X}}\right)$$

for both nontrivial twists. Orthogonality on H now gives equal main terms in its three residue classes. Therefore

$$\#\left\{ \begin{array}{l} n \leq X : n \equiv 1 \pmod{9}, \\ p \mid n \Rightarrow p \equiv 1 \pmod{3} \end{array} \right\} \sim \frac{G_3(1)}{3\sqrt{\pi}} \frac{X}{\sqrt{\log X}}. \quad (4.8)$$

There are $O(X^{1/3})$ perfect cubes up to X , so excluding cubes and the value 1 does not affect the main term. We have proved

$$\#\{a \in \mathcal{A} : a \leq X\} \sim \frac{G_3(1)}{3\sqrt{\pi}} \frac{X}{\sqrt{\log X}}. \quad (4.9)$$

Substituting $X = B/32$ by (4.3) proves Theorem 1.2.

4.2 A clean subfamily

Before returning to the prime line, it is useful to isolate a simpler, slightly thinner multiplicative subfamily

$$\mathcal{A}_9 = \{a > 1 : a \notin \mathbb{Q}^3, p \mid a \Rightarrow p \equiv 1 \pmod{9}\}. \quad (4.10)$$

It is obtained from the semigroup generated by primes $1 \pmod{9}$ by removing the perfect cubes and 1, and it is contained in $\mathcal{A}$. Its Dirichlet series has the form

$$\mathcal{F}_9(s) = \prod_{p \equiv 1 \pmod{9}} (1 - p^{-s})^{-1} = \zeta(s)^{1/6} G_9(s), \quad (4.11)$$

where G_9 is holomorphic and nonzero near $s = 1$. Equivalently,

$$G_9(1) = \lim_{s \rightarrow 1^+} (s - 1)^{1/6} \prod_{p \equiv 1 \pmod{9}} (1 - p^{-s})^{-1} > 0. \quad (4.12)$$

Another application of Selberg–Delange gives

$$\#\{a \in \mathcal{A}_9 : a \leq X\} \sim \frac{G_9(1)}{\Gamma(1/6)} \frac{X}{(\log X)^{5/6}}. \quad (4.13)$$

The full family is larger because it may combine primes in all three classes $1, 4, 7 \pmod{9}$, subject only to their product being $1 \pmod{9}$.

4.3 Comparison with the prime subfamily

Restricting to primes $\ell \equiv 1 \pmod{9}$, the prime number theorem in arithmetic progressions gives

$$\#\{\ell : H(y_\ell) \leq B\} = \pi(B/32; 9, 1) \sim \frac{B}{192 \log B}. \quad (4.14)$$

Thus the multiplicative family improves the prime construction by a factor of order $\sqrt{\log B}$. This improvement is arithmetic rather than geometric: all targets still lie on the same rational line. The line was the geometric discovery; recognizing that its local criterion is multiplicative is what changes the counting exponent. The pleasure is that the same local proof contains both facts once it is read in the right order.

The analytic input used above is the classical Selberg–Delange theorem in the $\zeta(s)^\rho G(s)$ form and its character-twisted version; we refer to de la Bretèche and Tenenbaum [2].

5 Optimality of degree five

The construction began with a quintic because the Berend–Bilu example is a product of degrees three and two. It is natural to ask whether the same fixed-map phenomenon could have been found in smaller degree. Here the arithmetic answers before the geometry does: a nontrivially intersective polynomial cannot have degree below five. The degree in which the example first appears is, pleasantly, also the first degree in which it could appear.

Define d_{HP} to be the least geometric degree of a polynomial Keller map over $\mathbb{Q}$ having a finite regular fiber that satisfies (1.1).

Theorem 5.1. *The minimal Hasse-failing geometric degree is*

$$\boxed{d_{\text{HP}} = 5}. \quad (5.1)$$

Proof. Let $F : \mathbb{A}_{\mathbb{Q}}^n \rightarrow \mathbb{A}_{\mathbb{Q}}^n$ be a Keller map of geometric degree d , and let $X = F^{-1}(y)$ be a finite regular fiber. Because F is étale, each geometric point of X extends, after shrinking around y , to a distinct finite étale branch. The geometric generic fiber has d points, so

$$\text{rank}_{\mathbb{Q}} X \leq d. \quad (5.2)$$

Suppose X is regular, everywhere locally soluble, and has no rational point. Choose an integral primitive element for its finite étale coordinate algebra. Thus

$$X \simeq \text{Spec } \mathbb{Q}[T]/(P) \quad (5.3)$$

for a monic squarefree polynomial $P \in \mathbb{Z}[T]$ of degree $\text{rank}_{\mathbb{Q}} X$. A $\mathbb{Q}_p$ -root of the monic polynomial is integral. Roots over every $\mathbb{Q}_p$ therefore give roots modulo every prime power, and the Chinese remainder theorem gives a root modulo every positive integer. The absence of a $\mathbb{Q}$ -point says that P has no rational root.

For completeness, the low-degree consequence of Berend and Bilu’s criterion can be seen directly from the possible irreducible-factor degrees [1]. With no rational root, a polynomial of total degree at most four is either irreducible of degree 2, 3, or 4, or is a product of two irreducible quadratics. In the irreducible case a derangement in the transitive Galois action gives a Frobenius class with no fixed root. In the two-quadratic case, if the quadratic fields agree then an inert Frobenius works for both factors; if they are distinct, their biquadratic compositum contains an element nontrivial in both quadratic quotients. The Berend–Bilu fixed-point criterion (equivalently, Chebotarev applied to these classes) then supplies a prime modulo which there is no root. Hence no polynomial of degree at most four can have a root modulo every positive integer without already having a rational root. Thus

$$5 \leq \text{rank}_{\mathbb{Q}} X \leq d. \quad (5.4)$$

On the other hand, Theorem 1.1 gives a full Hasse-failing fiber of a degree-five Keller map. Hence $d_{\text{HP}} = 5$. $\square$

Remark 5.2. *The lower bound is arithmetic and does not depend on the quadratic-gauge construction. The upper bound is stronger than mere realizability: one fixed degree-five map attains the minimum at infinitely many targets.*

6 Exact verification and finite data

The proof has three different assurance layers. The determinant, inverse factorization, resultants, heights, and finite counts have exact symbolic or integer checks. The algebraic fiber argument and the uniform local theorem also have a Lean formalization. The Selberg–Delange asymptotic and the analytic input to degree minimality remain ordinary mathematical proofs. These layers should not be confused: in particular, the bounded count is not evidence from which the asymptotic is inferred.

Run

$$\begin{aligned} &.\text{venv/bin/python scripts/verify_infinite_hasse_keller_fibers.py}, \\ &.\text{venv/bin/python scripts/verify_multiplicative_hasse_artifact.py}. \end{aligned} \quad (6.1)$$

The first command expands F_0 , verifies $\det DF_0 = -2$, checks the target-line factorization (2.18), the factor discriminants and resultant, the 3-adic substitution, and the first prime and composite parameters. The determinant-one identity for Φ then follows from the two displayed linear transformations in (2.4).

The second command uses only the Python standard library. It checks support conditions, cube exclusion, target primitivity, the height formula, and stable digests of the enumerated parameter sets. At $X = 10^6$ it gives

family	count	scaled count
$\mathcal{A}$	26846	$\#\mathcal{A}(X)\sqrt{\log X}/X = 0.0997845$
$\mathcal{A}_9$	16107	$\#\mathcal{A}_9(X)(\log X)^{5/6}/X = 0.143655$
primes $1 \pmod{9}$	13063	
products of two such primes	2853	

(6.2)

The generated JSON certificate has SHA-256 hash

$$\text{d4cbaa54f2a3f436f21e26b17e24ebdd8894dc7877f14e2caa9aeaf603f81cb9.} \quad (6.3)$$

The machine-checked theorem layer is built separately by

$$\begin{aligned} &\text{cd formal/finite-etale-keller,} \\ &\text{lake build FiniteEtaleKeller.FixedHassePaperCertificate.} \end{aligned} \quad (6.4)$$

The machine-checked chain consists of

`FixedHasseFamily.lean`, `FixedHasseMap.lean`, `FixedHasseLocal.lean`,
`FixedHasseHeight.lean`, and `FixedHasseArithmetic.lean`, collected by the isolated publication entry
`point FixedHassePaperCertificate.lean`.

These modules prove, respectively, the centered factorization and squarefreeness certificates, the exact determinant-one normalization and complete literal-fiber equivalence, the uniform prime-by-prime local theorem, and the primitive coordinates with exact height $32a$, followed by multiplicative closure of the congruence/support conditions and the complete prime-progression certificate. In particular, Lean proves that a prime $\ell \equiv 1 \pmod{9}$ cannot be a rational cube. The local module derives oddness from prime support, so its final declaration `paperFiberPoint_hasse_certificate` states directly that every admissible noncube parameter gives a literal fiber with no rational point, a real point, and a point over every $\mathbb{Q}_p$. The certificate contains no `sorry` and uses only Lean’s standard foundational axioms.

The exact displayed-map degree is included as well. Lean proves the polynomial identity obtained by applying $A^{-1}(x, y, z) = (2x, y, z)$ and $L^{-1} = L$ to Φ , recovering F_0 exactly. These source and target polynomial automorphisms preserve function-field degree, so the certified degree-five computation for F_0 is the geometric degree of the displayed map Φ , not merely of an unnormalized auxiliary map.

The table is a convergence and regression check. The local theorem applies to every parameter satisfying (1.8), while the asymptotics follow from the Euler products and Selberg–Delange; neither conclusion is inferred from the bounded enumeration.

7 Discussion

The prescribed-fiber theorem made the first Hasse example almost automatic: start with an intersective polynomial and compile its finite étale algebra into a full Keller fiber. That flexibility also concealed the next question. If the construction is allowed to absorb the polynomial into the map, it says nothing about how many arithmetic behaviors one map can carry.

The present argument began when those two roles were separated. The map should remember only the fixed quintic shape; the target should remember the parameter. The quantifier change is small to write and substantial in effect:

	map	fiber algebra	
prescribed fibers	varies	arbitrary of rank ≥ 3	(7.1)
this paper	fixed	$\mathbb{Q}[X]/((X^3 - a)(X^2 + X + 1))$, $a \in \mathcal{A}$.	

Two observations make the fixed row possible. The first is geometric: after the translation $X = S - \frac{1}{2}$, the quadratic $X^2 + X + 1$ becomes $S^2 + \frac{3}{4}$. The parameter a then moves only the quadratic and constant coefficients, which are exactly the two free coefficients of the quadratic-gauge target pencil. The second is arithmetic: once the local proof is written place by place, primality is nowhere essential. The congruence-and-support core is multiplicatively closed; removing its perfect cubes leaves the admissible family. The count grows from the prime scale $B/\log B$ to $B/\sqrt{\log B}$.

There is also a useful warning in the construction. The target line is exceptional from the viewpoint of generic monodromy. Every inverse polynomial on it has the same quadratic-times-cubic factorization. This is not an accidental defect: the factorization is what lets one component supply local points precisely where the other does not. The deterministic local behavior on the line therefore says little about the distribution of local points on the ambient three-dimensional target.

The exact scope is equally important. Theorem 1.2 counts one sufficient multiplicative family. It does not classify all Hasse failures on the line, and still less all Hasse-failing fibers of the map. A converse local analysis would have to decide which arbitrary rational parameters make at least one of the two factors soluble at every place. Integral source points introduce another layer: a local root of the inverse polynomial need not reconstruct to an integral source point without further valuation control.

What the paper does establish is a clean progression. What began with one stubborn quintic ends with a more generous picture: the classical example is the first visible point on a rational target line belonging to one Jacobian-one map of the smallest possible geometric degree. A half-unit translation reveals the line, a place-by-place proof reveals the multiplicative family, and the arithmetic itself explains why degree five is minimal. These are small observations individually, but together they turn an example into a landscape.

Provenance and AI assistance. The arithmetic polynomial at $a = 19$ is due to Berend and Bilu [1]. The fixed-map construction is obtained by specializing the quadratic-gauge prescribed-fiber mechanism of [4].

I used generative AI tools during exploration, symbolic-checker development, proof review, literature search, and manuscript editing. I checked the stated mathematical claims and take responsibility for the paper.

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